HRR (Hb / RDW) → haptoglobin-mediated heme-detoxification failure → NRF2/BACH1-HMOX1 → ferroptosis after SAH

bulk-transcriptome · WGCNA · mouse brain single-cell · deconvolution · SCENIC(NRF2/BACH1) · CellChat-style · FerrDb · validated MR/coloc engines

1 CLINICAL + SYSTEMIC EVIDENCE

MIMIC-IV n = 1,161 aSAH
HRR = Hb / RDW independently predicts in-hospital death

Hb arm → O2 delivery RDW arm → RBC turnover

Acute blood GSE36791 (43 vs 18)
23 up / 12 down DEGs (FDR<.05): HP · HMOX1· FTH1 SLC40A1 · SLC25A37 · NFE2 · MARC1 + inflammation

32 candidate genes ∩ red-cell sig.
Erythrocyte diff. p≈5e-26 · GSEA: NF-κB/ROS/hypoxia/ferroptosis

WGCNA (signed)
SAH module eigengene r=0.55 (p=4e-06)
heme-iron genes HP/MARC1/SLC40A1/ACSL4/FTH1 high MM

Human-blood deconvolution
Innate↑ (neutrophil d=1.56) / adaptive↓ (T d=-1.92)
Heme-iron program up (Δ=0.55, p=4e-05)

Ferroptosis ~ heme across patients
Spearman ρ = 0.54 (p = 7e-06)

2 SINGLE-CELL MECHANISM (mouse brain GSE311852)

SAH red-cell lysis / hemolysis
free Hb released

HP haptoglobin scavenging
Neutrophil (Hp ↑, log2FC≈2)

Myeloid uptake CD163 / MRC1
scRNA: Cd163 <6% — MRC1 + S100-TLR4 engaged

Heme overload → NRF2↑ / BACH1↓
regulon: NRF2 activity↑, BACH1 activity↓ (microglia)

HMOX1 hyper-activation
Myeloid Hmox1 ↑ (log2FC≈2.4)

Fe2+ / ROS — HDFI (detox-failure index)
z(HMOX1)−z(FTH1+FTL)−z(GPX4); predicts neuron ferroptosis ρ=0.34

Ferroptosis
Neuron: Gpx4↓, ACSL4/SLC7A11/PTGS2↑; blood Exec p=0.003

Secondary neuronal injury
poor outcome (HRR link)

Temporal dynamics (Sham→24h→72h)

Neutrophils 0.5→38.6→3.7% (first wave)

Myeloid 14.8→32.3→53.3% (expansion)

Kruskal-Wallis all p < 1e-16

Upstream driver (SCENIC-lite)

NRF2 / NFE2L2 regulon ACTIVATED in SAH
(0.59 vs 0.28, p<1e-4) — not NFE2-p45;
BACH1 also up

ML transportability (XGBoost)

Heme-iron panel generalises best externally
(CV 0.80 → ext AUC 0.59) vs immune DEG 0.52

3 HONEST BOUNDARIES

- X Cd163 not induced (p=0.83); HP–CD163 fails CellChat (Cd163<6%) → MRC1/S100-TLR4
- X Bulk-blood heme/HP signal is COMPOSITION-driven (does not survive adjustment)
- X HDFI NS in blood (Δ+0.18, p=0.28) — signal is single-cell
- X DPT pseudotime & FerrDb Neu→Mac→Neuron chain NOT monotonic
- X External AUC < 0.75 — acute signal ≠ remote aSAH

WHAT PUBLIC DATA CANNOT PROVE — the author must generate:

(i) real MR / TWAS / coloc causal estimates (engines validated on synthetic truth: IVW β=0.40→0.41, coloc PP4=1.0; sumstats access-blocked) · (ii) plasma proteomics + MR-PheWAS · (iii) spatial co-localisation
(iv) the HRR → HDFI → brain-injury → death MEDIATION/SEM bridge on a matched cohort (engine validated: ACIE 0.48→0.481; no public HRR+plasma cohort here)
(v) an ACUTE human cohort with admission Hb/RDW + outcome · (vi) animal/cell validation: ferrostatin-1, HMOX1 inhibitor ± timing, CD163/MRC1 blockade